WORKED SOLUTION

STEP I 1994, Question 11

Mechanics • Connected Particles on Incline

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Read the question before you go on. It is in the STEP Support Programme's question database:

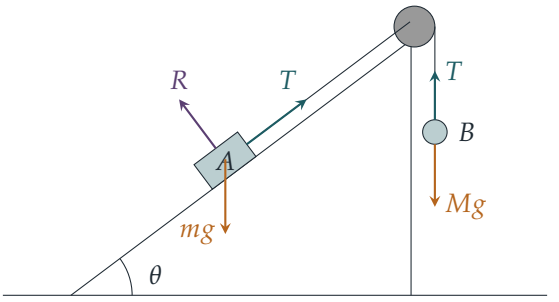
step.maths.org/questions/94-s1-q11

The key idea

Start with the geometry. The string is exactly as long as the track, so when the wagon is at the bottom the ball hangs right at the pulley. So the ball falls through the height of the incline, $d\sin\theta$, and the wagon climbs the same distance up the slope. While the string is taut the system moves with one constant acceleration. Once the ball lands, the string goes slack and the wagon just coasts up a smooth slope. So part (ii) asks whether its leftover speed carries it the rest of the way to the pulley.

Solution

We model the wagon and ball as particles, the track as smooth, and the string as light and inextensible over a light frictionless pulley. So the tension T is the same on both sides, and the wagon and ball have the same speed while the string is taut.



Part (i)

Let a be the acceleration of the ball downwards, which is also the acceleration of the wagon up the slope. Newton's second law for the ball (vertically) and for the wagon

(along the slope) gives

$$\begin{aligned} Mg - T &= Ma, \\ T - mg\sin\theta &= ma. \end{aligned}$$

The normal reaction R is perpendicular to the slope and does not enter. Adding the two equations eliminates T:

$$a = \frac{(M - m\sin\theta)\, g}{M + m}.$$

The system starts from rest, so the ball moves at all only if $a > 0$. (If $a \leq 0$ nothing happens: with $a = 0$ the system stays in equilibrium, and the string can't push, so $a < 0$ is impossible.) So the ball falls to the ground if and only if

$$M > m\sin\theta.$$

When the wagon is at the bottom of the track the full length d of the string lies along the slope, so the ball starts level with the pulley, at height $d\sin\theta$ above the ground. It falls this distance from rest with constant acceleration a, so by $v^2 = u^2 + 2as$,

$$v^2 = 2 \cdot \frac{(M - m\sin\theta)g}{M + m} \cdot d\sin\theta = \frac{2g(M - m\sin\theta)\, d\sin\theta}{M + m},$$

as required.

Part (ii)

When the ball hits the ground, the wagon has moved $d\sin\theta$ up the track and is moving at speed v. That leaves a distance

$$d - d\sin\theta = d(1 - \sin\theta)$$

to the pulley.

Ask yourself What happens to the string once the ball has landed?

The ball stops, but the wagon is still moving up the slope, towards the pulley. So the string goes slack and $T = 0$. The only force along the slope is now the component of gravity, so the wagon decelerates at $g\sin\theta$. It comes to rest after a further distance s given by $0 = v^2 - 2g\sin\theta\, s$:

$$s = \frac{v^2}{2g\sin\theta} = \frac{(M - m\sin\theta)\, d}{M + m}.$$

(The wagon then rolls back down, and nothing more happens near the pulley.)

The wagon doesn't collide with the pulley if it stops before reaching it, that is, if $s < d(1 - \sin\theta)$. Since $d > 0$ and $M + m > 0$, this is equivalent to

$$\begin{aligned} M - m\sin\theta &< (M + m)(1 - \sin\theta) \\ &= M + m - M\sin\theta - m\sin\theta, \end{aligned}$$

which simplifies to $M\sin\theta < m$.

Combining this with part (i), the ball reaches the ground and the wagon stops short of the pulley exactly when

$$m\sin\theta < M < \frac{m}{\sin\theta}.$$

If $M\sin\theta = m$, the wagon just reaches the pulley with zero speed. That's a touch, not a collision.

Checks

- The furthest the wagon gets up the track is $d\sin\theta + s = \dfrac{Md(1+\sin\theta)}{M+m}$. This is less than d exactly when $M\sin\theta < m$, which agrees with the condition above. As $m \to 0$ it tends to $d(1+\sin\theta) > d$: a weightless wagon always hits the pulley, as you'd expect.
- At $\theta = 90°$ the condition $M < m$ contradicts $M > m\sin\theta = m$ from part (i). That's right: on a vertical track the wagon starts at the bottom directly under the pulley, the ball starts at the top, and the wagon rises exactly as far as the ball falls, which is all the way to the pulley.
- Numerically, with $m = 2$, $M = 1.5$, $\theta = 30°$, $d = 3$ and $g = 9.8$, the formula gives $v^2 = 4.2$ and a furthest point of $\frac{27}{14} \approx 1.93 < 3$. Stepping the motion forward in small time steps gives the same values. Random cases also confirm that the collision happens exactly when $M\sin\theta > m$.

Common slips

- Taking the ball's drop to be d rather than $d\sin\theta$. The ball starts at the pulley, which is at height $d\sin\theta$.
- Assuming the string stays taut after the ball lands, or forgetting the coasting phase altogether and comparing only $d\sin\theta$ with d.
- Using g instead of $g\sin\theta$ for the wagon's deceleration once the string is slack.

Beyond STEP

The question splits the motion into phases with different constraints: string taut, then string slack. This is the simplest example of a *unilateral constraint*, which can pull but not push. Such constraints come up throughout mechanics, from chains falling off tables to contact problems in robotics. Energy methods give a quick alternative here: in the coasting phase, the wagon's kinetic energy $\frac{1}{2}mv^2$ turns into potential energy $mgs\sin\theta$. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both viewpoints carefully.

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